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lar chains has a great influence on the mechanical properties.

Thermosets have higher stiffness, strength, and creep resistance compared to thermoplastics because the molecular chains are interconnected by cross-links. These cross-links resist chain slippage under tension and therefore increase mechanical properties.

Epoxy resin is the most widely used polymer for the aerospace industry and finds applications in carbon fibre composites and as a structural adhesive. Epoxy resin has good stiffness, strength, and environmental durability and can be cured at moderate temperatures (below 180°C) with little shrinkage.

Other thermosets used in composite materials include bismaleimide and polyimide, which are suitable for high temperature applications. Phenolic resin has excellent resistance to flammability and is used in car interior composite and moulded parts.

Thermoplastics have high toughness, ductility, and impact resistance, but high processing cost and low creep resistance limit their use in the aerospace industry. Thermoplastics such as PEEK are used as a matrix phase in fibre composites that require impact resistance. Polycarbonate is used for aircraft windows due to its high transparency, scratch resistance, and impact resistance.

Elastomers lack the stiffness and strength necessary for aerospace applications, although their high elasticity makes them useful as a sealing material. Elastomers have high elasticity due to their coiled molecular structure that responds under load, similar to a spring.

The adhesives used to bond aircraft components and sandwich composites are often thermosetting, such as epoxy resin. Adhesives should have low shrinkage during cure to avoid residual tensile stress build-up at the joint. Proper surface preparation is essential to ensure a long-lasting, high-strength bond with a polymer adhesive.

The mechanical properties of polymers are inferior to those of aerospace structural metals and their use is restricted to relatively low temperature applications. Polymers transform from a hard, glassy state to a soft, rubbery state during heating, and the maximum working temperature is defined by the thermal deflection temperature or the glass transition temperature.

Polymers often contain additives for special functions such as colour, toughness, or fire. Radar absorbent materials are a special kind of polymer that converts radar (electromagnetic) energy into some other form of energy (such as heat) and thus improves the stealth of military aircraft. RAM should be used with other stealth technologies, such as aircraft shape design adaptations, to minimise radar cross-section. EFY

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